

How is the association between a PhD student's gender and their choice of thesis topic in computer science at TU Berlin?

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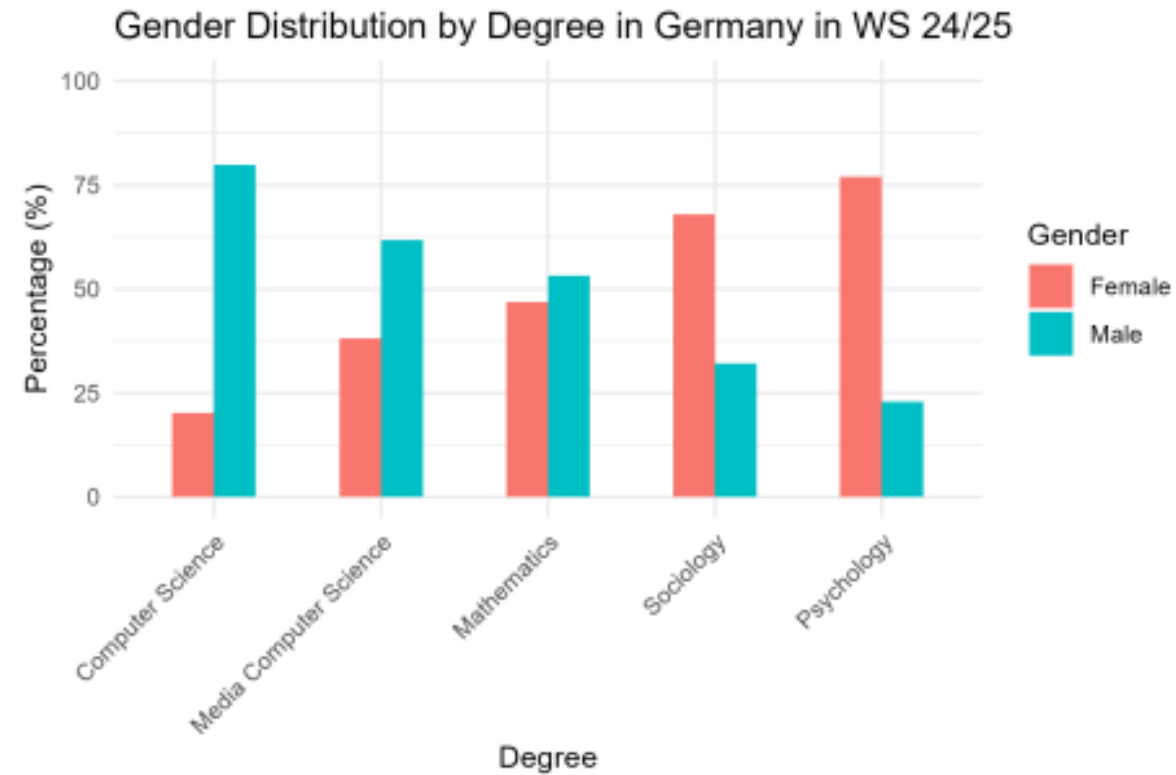
- Motivation & Background
- Research Question
- The Dataset
- Methods: current state and next steps
- References
- Appendix

Motivation

19% of computer science,
15% of mechanical engineering,
19% of electrical engineering students at TUB are female.

47% of media computer sciences students at TUB are female.

Motivation

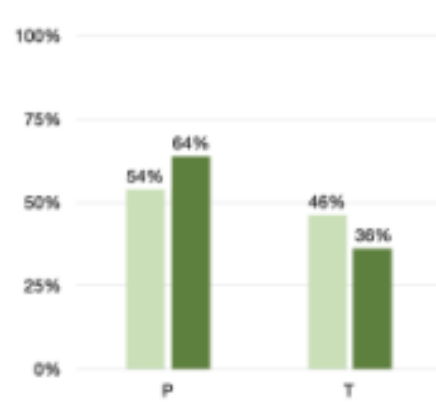


Background – People Over Things

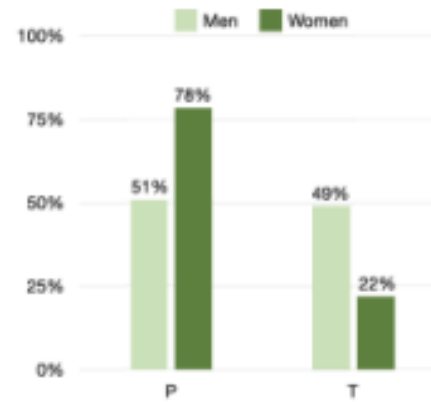
Gender has an impact on peoples wish to work with people or things.

- Study in the Introductory Programming course at the IT University of Copenhagen
- 152 first semester students (37 female, 115 male)
- Choosing in 4 mandatory assignments between 2 isomorphic people and thing related assignments
 - «A hotel employee wants to have a visual overview.»
 - «A screen has to give a visual overview»

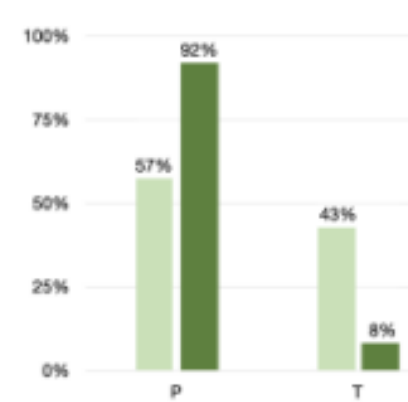
Background – People Over Things



(a) Assignment B.



(b) Assignment C.



(c) Assignment D.

Research Question

How is the association between a PhD student's gender and their choice of thesis topic in computer science at TU Berlin?

Acknowledgement:

Gender is not a binary construct, but for the purposes of this thesis and the restrictions of the dataset I will be using the assigned gender of the students: male and female.

The Dataset

The dataset that shows doctoral degrees at Faculty IV at TU Berlin from 2000 to 2025.

Initially -> 1860 rows, 25 columns

Most important:

- Gender (1574 male, 261 female graduates)
- Title of thesis
- Finishing year
- Degree (112 different entries)

The Dataset

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**PROBLEM: CAN'T LIMIT THE RESEARCH QUESTION TO
COMPUTER SCIENCE**

Methods – Sentence Transformers (SBERT)

- Built on BERT -> model for natural language processing, analyzing text in both directions and understanding context of words
- Transformer based encoder architecture – take input sequence, transform into numerical embeddings for each token
- Sentences -> SBERT, embeddings for entire sentences
- Semantic similarity? -> compare vectors: similarity measures

Methods - Clustering

- Clustering with
 - K-Means -> knowing number of clusters beforehand
 - HDBSCAN
 - > no labels created
- Need to give names to clusters in natural language

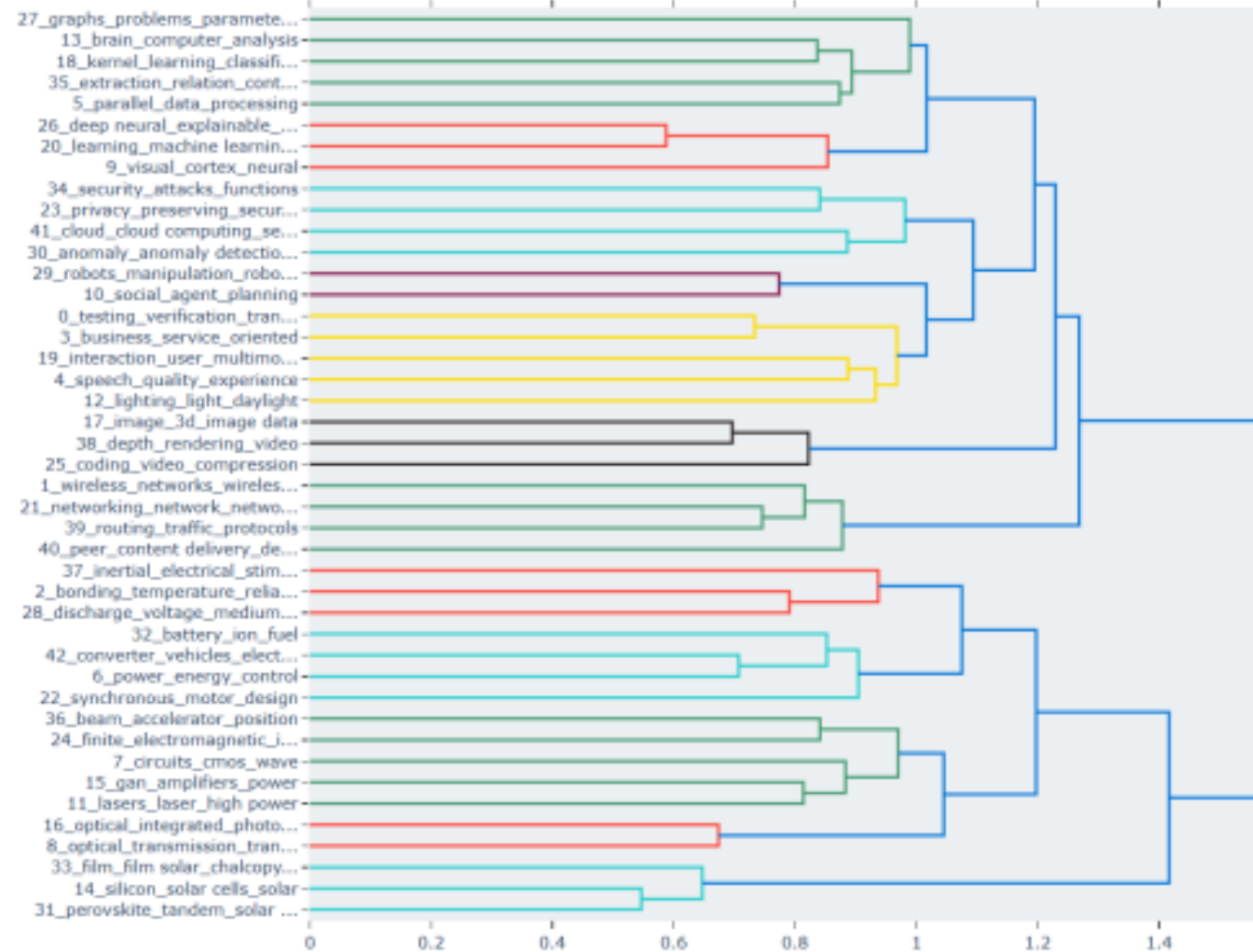
Methods - BERTopic

- Python library that generates document embeddings with pre-trained transformer-based language models, clusters them, provides topic representations

Methods – BERTopic, Applied

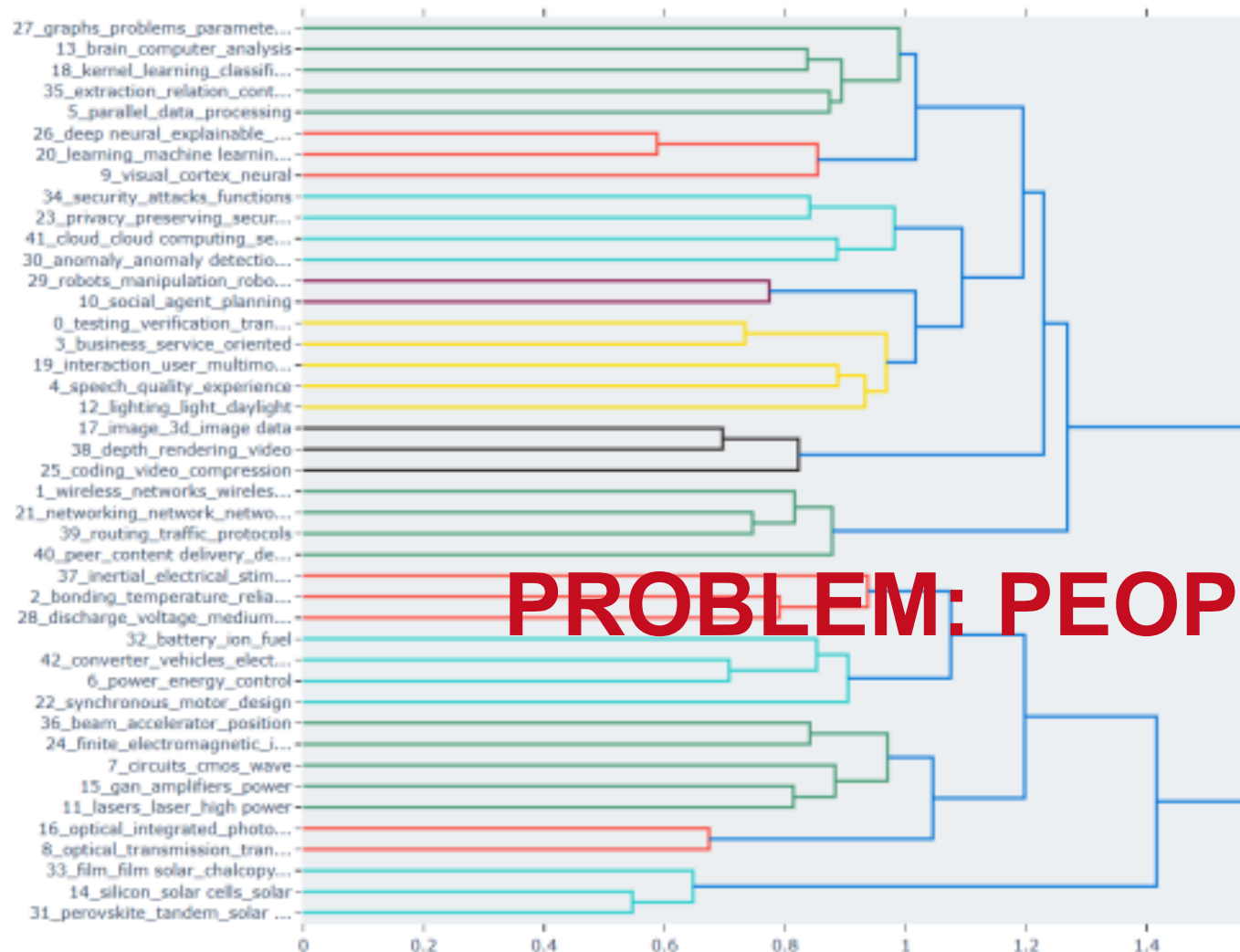
- Translated every thesis title to English using Pandas and deep-translator
- Embedding model: all-mpnet-base-v2
- Topic model: BERTopic
 - Each group: topic_id
 - get_topic function: keywords of topic groups

Hierarchical Clustering



Hierarchical
clustering
dendrogram
– 44 groups

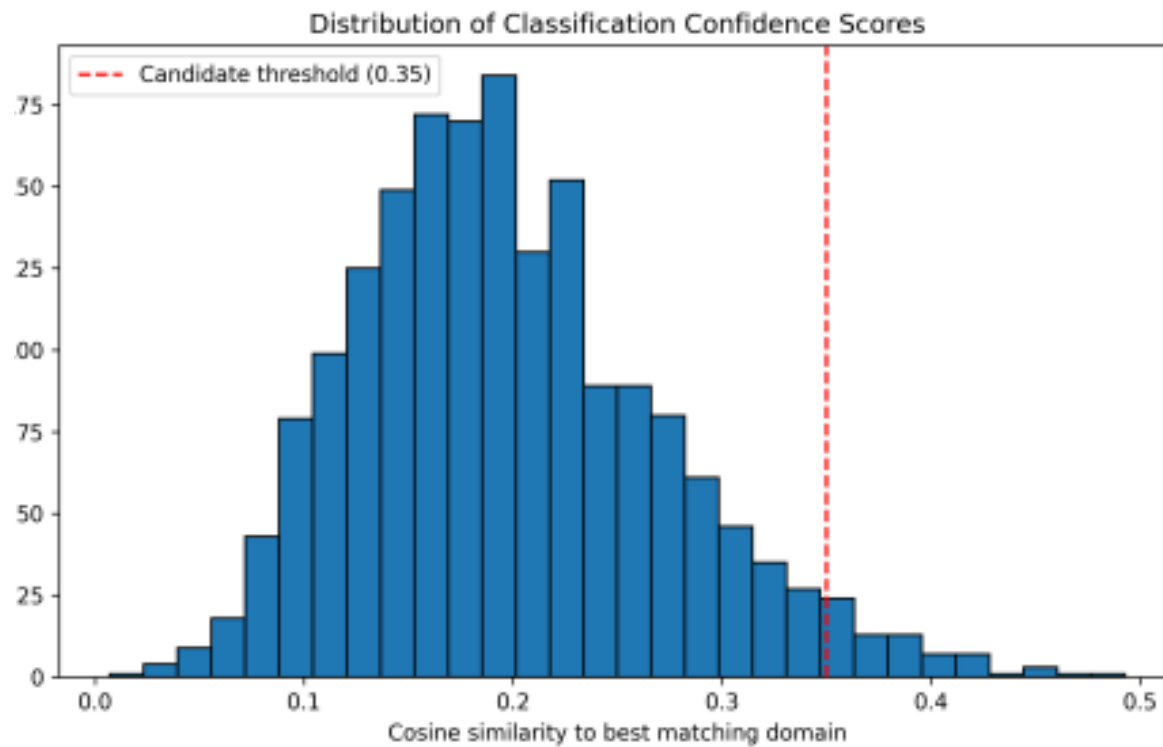
Hierarchical Clustering



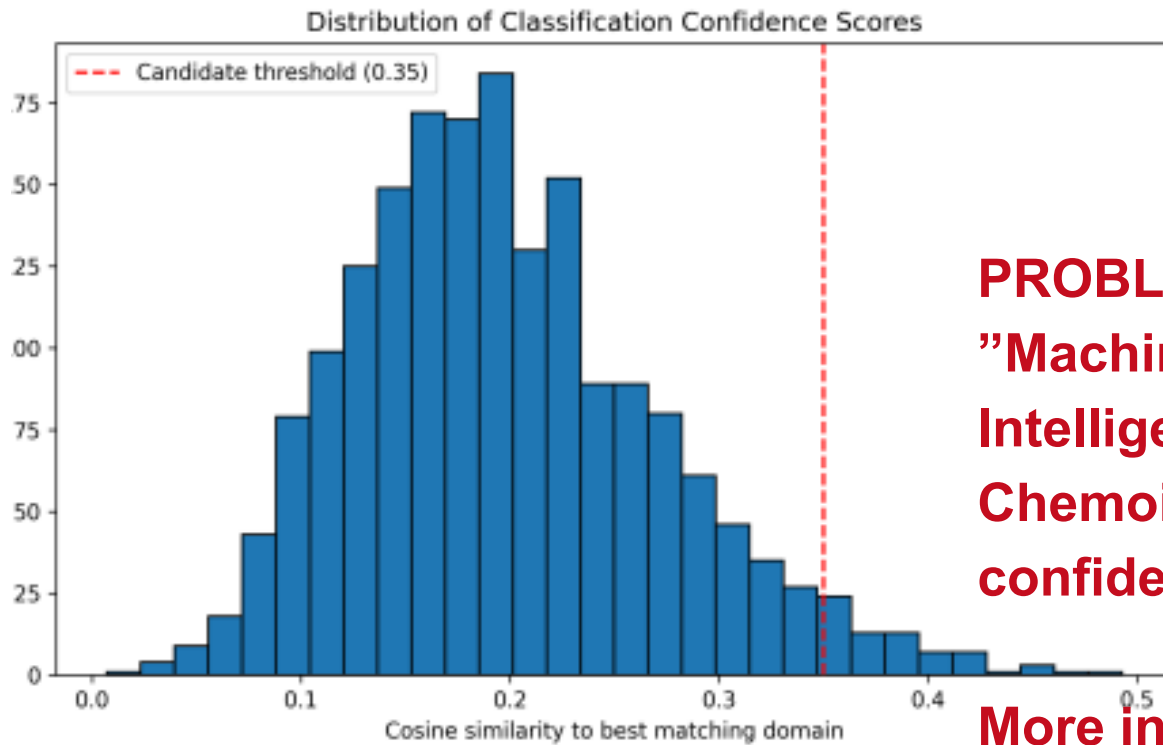
Hierarchical
clustering
dendrogram
– 44 groups

Methods – SBERT

- ChatGPT -> create 20 meaningful cluster names and short labels for better visibility
 - ”Research in economics, markets, finance, business, and organizational behavior”,
 - ”Research in computer science, artificial intelligence, data systems, and computational methods”,
 - ”Research in electrical engineering, electronics, telecommunications, and signal processing”...
- Embedding model: all-mpnet-base-v2
- Encode titles and groups
- Cosine similarity



- 0,35 candidate treshold, random
- Looking at matches with high confidence scores:



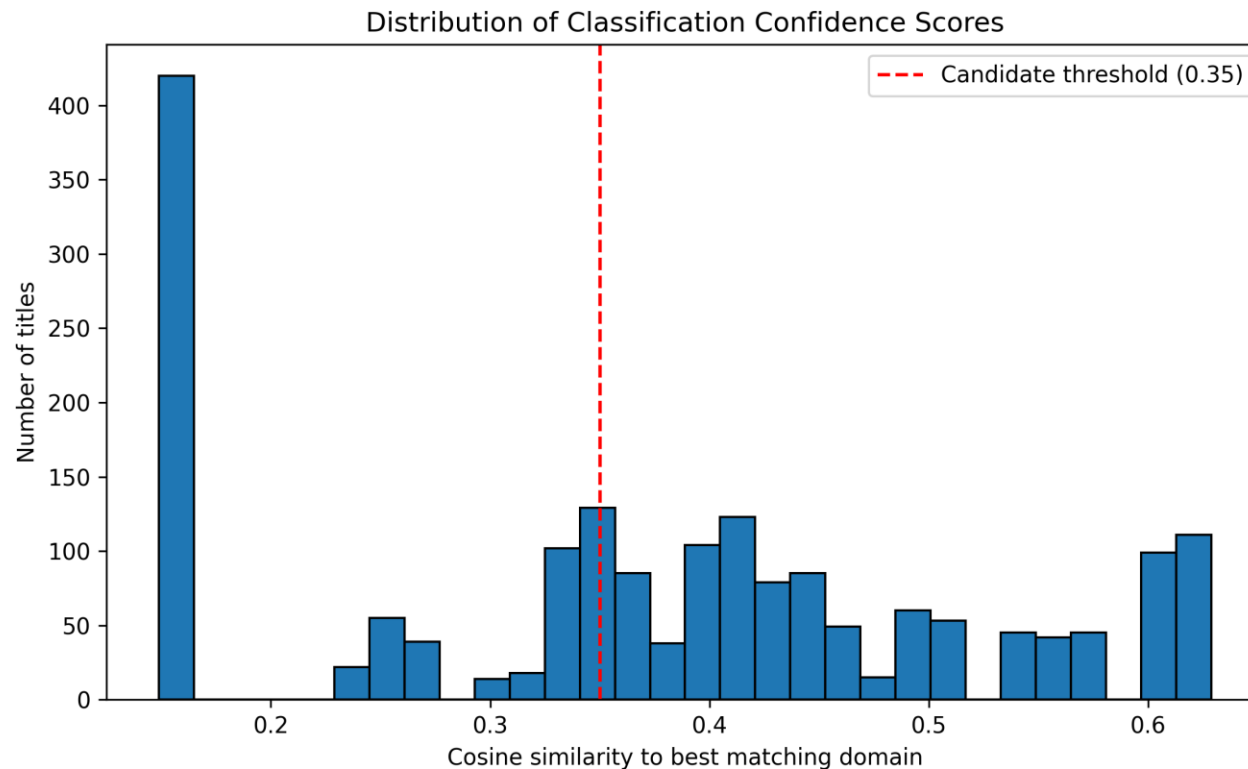
- 0,35 candidate treshold random
- Looking at matches with high confidence scores:

PROBLEM:

"Machine Learning Methods for Life Science: Intelligent Data Analysis in Bio- and Chemoinformatics" - "Computer Science & AI" - confidence score 0,414

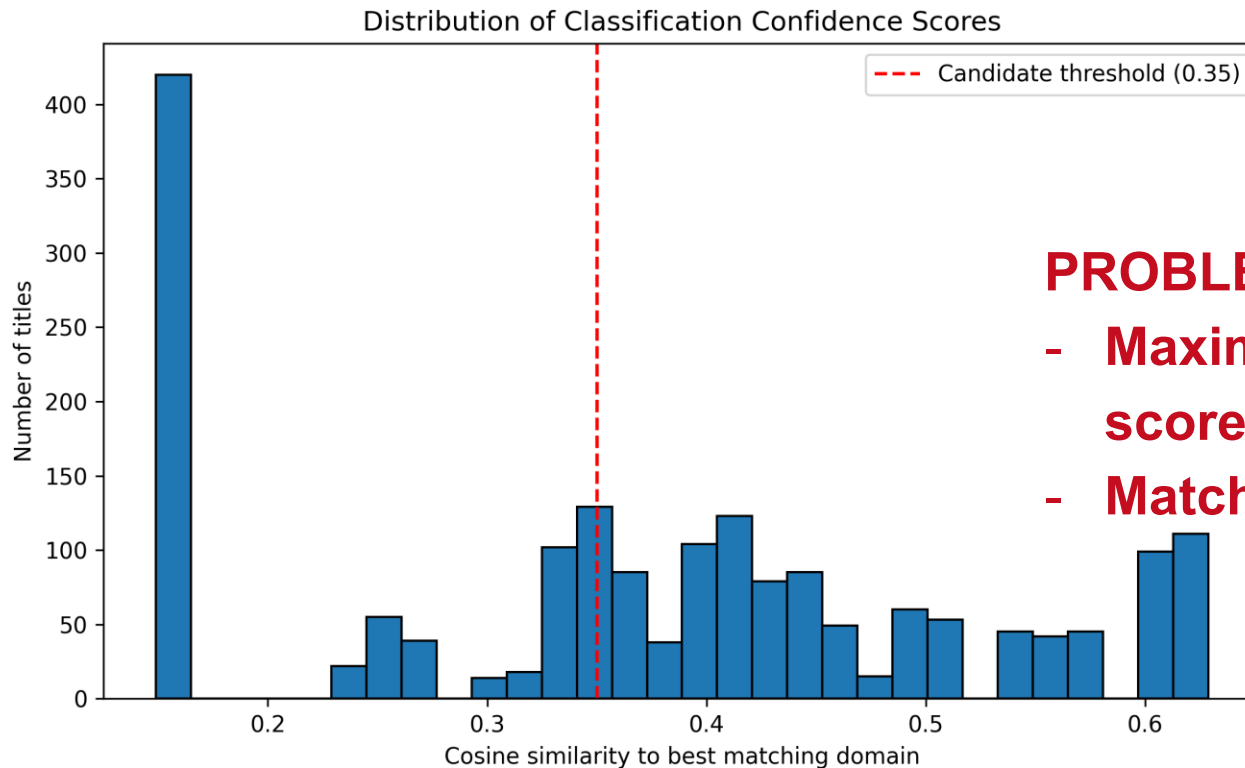
More insightful if medicine and/or biology also

Methods – SBERT, more groups



- 30 topic names instead of 20

Methods – SBERT, more groups



- 30 topic names instead of 20

PROBLEM:

- Maximum amount of matches in confidence score 0,1: too low
- Matches very scattered, hard to understanding

PROBLEM: PEOPLE/THING CHOICE IS ALSO ABOUT HOW A QUESTION WAS PUT.

Methods – Next

- SciBERT: trained on scientific text from the corpus of semanticscholar.org
- Specter2: converts titles and abstracts of scientific papers into numerical embeddings.
- Categorization into people/thing related topics
- Looking at the frequency of each topic for the genders and comparing them within and across
- Visualizations
- Discussion

THANK YOU.



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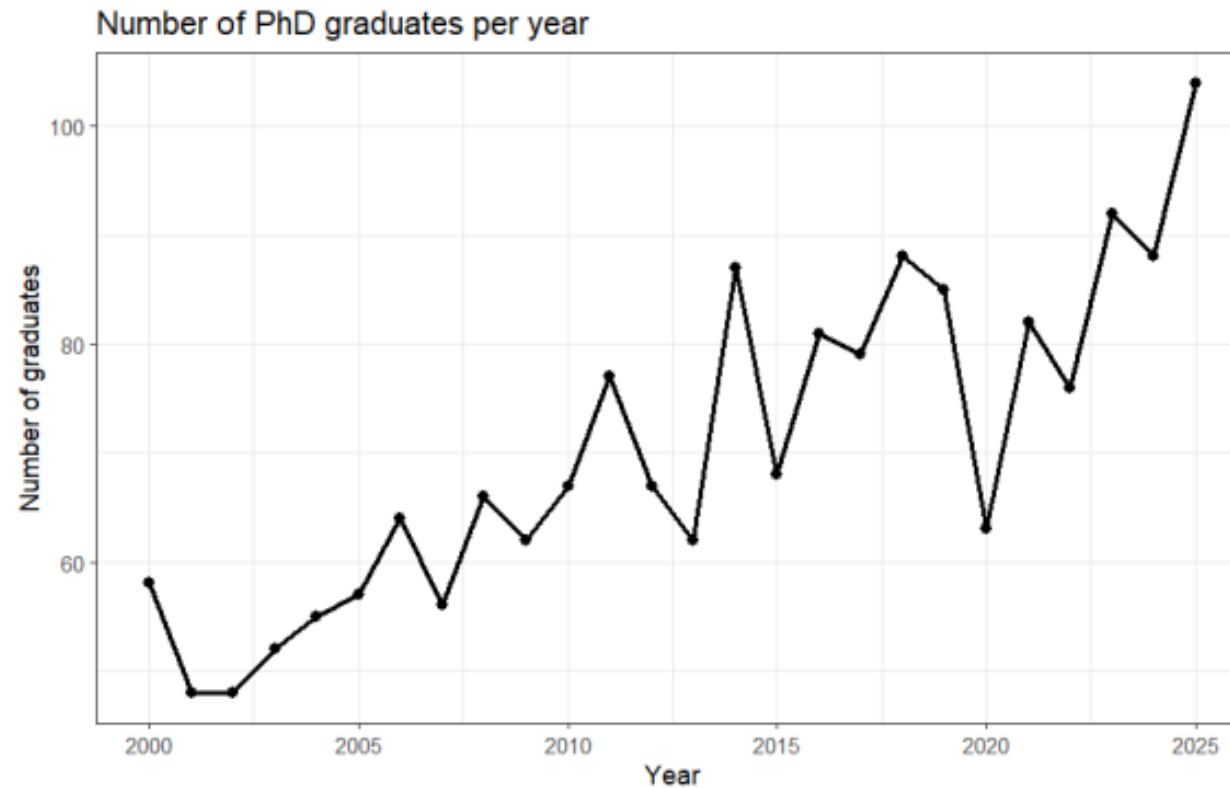
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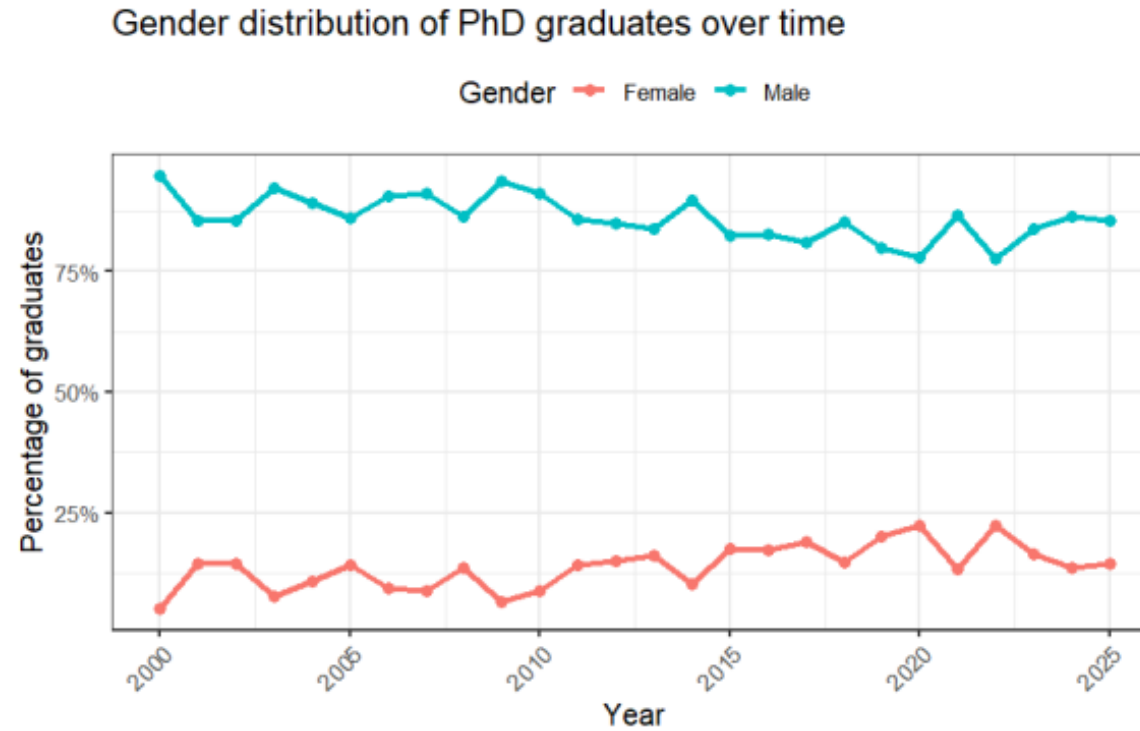
Appendix

Number of PhD graduates per year



Appendix

Gender distribution of PhD graduates over time



Appendix

Dataset column names – full

Nr. (number) - Name (surname - deleted for anonymization) - Vorname (name - deleted for anonymization) - m/w (male/female) - dt/Ausl. (German/foreigner) - Grad (degree) TU-Absolvent*in (TU-graduate) - Fak-Absolvent*in (faculty-graduate) - FH-Master (Fachhochschule master) - Dr. (Dr.) - Kooperative Promotion mit FH (Joint doctoral program with a Fachhochschule) - kumulativ (cumulative) - Titel der Dissertation (Titel of thesis) - Erstberichterstatter ... Sechstberichterstatter (First reporter ... sixth reporter) - WM-Z FG/Dauer (research staff - fixed-term contract / permanent contract) - Drittm/Stip FG/Dauer () Datum Aussprache (date) - Gesamtnote (final note) - Alter bei Abschluss der Promotion (Age at the time of completing the PhD) - Dauer der Promotion in Jahren (Duration of doctoral program in years)

Appendix

Teachers, mathematics and gender

- Teachers see the abilities of their students differently based on gender.

Appendix

Abbreviations

BERT -> Bidirectional Encoder Representation from Transformers

BERTopic:

- Embeds the text and does dimension reduction, clusters similar texts together, vectorizes clusters, calculates importance scores identifying key terms